MDTS4214_737_6

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Problem Set 6: Model Selection

Consider the "Hitters" data. For loading the data you need to install ISLR library. We wish to predict a baseball player's salary on the basis of various predictors. So first we need to choose an optimal model.

```
library(ISLR)
## ??Hitters
original_data=Hitters
```

Information on the data set:

A data frame with 322 observations of major league players on the following 20 variables.

*(i)*AtBat:Number of times at bat in 1986

*(ii)*Hits:Number of hits in 1986

*(iii)*HmRun:Number of home runs in 1986

*(iv)*Runs:Number of runs in 1986

*(v)*RBI:Number of runs batted in in 1986

*(vi)*Walks:Number of walks in 1986

*(vii)*Years:Number of years in the major leagues

*(viii)*CAtBat:Number of times at bat during his career

*(ix)*CHits:Number of hits during his career

*(x)*CHmRun:Number of home runs during his career

*(xi)*CRuns:Number of runs during his career

*(xii)*CRBI:Number of runs batted in during his career

*(xiii)*CWalks:Number of walks during his career

*(xiv)*League:A factor with levels A and N indicating player's league at the end of 1986

*(xv)*Division:A factor with levels E and W indicating player's division at the end of 1986

*(xvi)*PutOuts:Number of put outs in 1986

*(xvii)*Assists:Number of assists in 1986

*(xviii)*Errors:Number of errors in 1986

*(xix)*Salary:1987 annual salary on opening day in thousands of dollars

*(xx)*NewLeague:A factor with levels A and N indicating player's league at the beginning of 1987

Here, 'Salary' will be our Response variable.

1. Clean the data by removing all rows that have missing value in any variable. You will be working with this clean data.

```
cleaned_data=na.omit(original_data)
nrow_cleaned_data=dim(cleaned_data)[1][[1]];nrow_cleaned_data
```

```
## [1] 263
```

Comment: So the actual data size is 263.

\textbf{2. Split the data randomly into train-test keeping 80% as training sample.} and remaining as test sample.

```
train_data_size=trunc(0.8*263);train_data_size
```

```
## [1] 210
```

Comment: So, our training sample size is 210.

```
test_data_size=263-210;test_data_size
```

```
## [1] 53
```

Comment: So, our test sample size is 53.

```
# Ids of full data size
id=1:263

# 80% sample on ids of original data size for training sample
set.seed(123)
sample_id=sample(id,210)

# Training data
train_data=cleaned_data[sample_id,]

#Test data
test_data=cleaned_data[-sample_id,]
```

3. Perform the best subset selection on the training sample. While choosing the best model fit up to a 19 variable model. Choose the best model according to adjusted R squared, BIC , AIC and Cp values. In case the best model differs according to the selection criteria, report the best fitted model according to each selection criterion.

```
## Best Subset Selection on the Training Data:

#leaps package required
library(leaps)
```

```
## Warning: package 'leaps' was built under R version 4.5.3
```

```r
# regsubsets() with default method of Best Subset Selection
# nvmax fits the maximum no. of predictors considered in the full model
reg1=regsubsets(Salary~.,data=train_data,nvmax=19);reg1
```

```
## Subset selection object
## Call: regsubsets.formula(Salary ~ ., data = train_data, nvmax = 19)
## 19 Variables  (and intercept)
##            Forced in Forced out
## AtBat          FALSE      FALSE
## Hits           FALSE      FALSE
## HmRun          FALSE      FALSE
## Runs           FALSE      FALSE
## RBI            FALSE      FALSE
## Walks          FALSE      FALSE
## Years          FALSE      FALSE
## CAtBat         FALSE      FALSE
## CHits          FALSE      FALSE
## CHmRun         FALSE      FALSE
## CRuns          FALSE      FALSE
## CRBI           FALSE      FALSE
## CWalks         FALSE      FALSE
## LeagueN        FALSE      FALSE
## DivisionW      FALSE      FALSE
## PutOuts        FALSE      FALSE
## Assists        FALSE      FALSE
## Errors         FALSE      FALSE
## NewLeagueN     FALSE      FALSE
## 1 subsets of each size up to 19
## Selection Algorithm: exhaustive
```

```r
#(*) in summary indicate the predictors in the best fitted subset model.
reg1_s=summary(reg1);reg1_s
```

```
## Subset selection object
## Call: regsubsets.formula(Salary ~ ., data = train_data, nvmax = 19)
## 19 Variables  (and intercept)
##            Forced in Forced out
## AtBat          FALSE      FALSE
## Hits           FALSE      FALSE
## HmRun          FALSE      FALSE
## Runs           FALSE      FALSE
## RBI            FALSE      FALSE
## Walks          FALSE      FALSE
## Years          FALSE      FALSE
## CAtBat         FALSE      FALSE
## CHits          FALSE      FALSE
## CHmRun         FALSE      FALSE
## CRuns          FALSE      FALSE
## CRBI           FALSE      FALSE
## CWalks         FALSE      FALSE
## LeagueN        FALSE      FALSE
```

```
## DivisionW      FALSE      FALSE
## PutOuts        FALSE      FALSE
## Assists        FALSE      FALSE
## Errors         FALSE      FALSE
## NewLeagueN     FALSE      FALSE
## 1 subsets of each size up to 19
## Selection Algorithm: exhaustive
##           AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits CHmRun CRuns CRBI
## 1  ( 1 )  " "   " "  " "   " "  " " " "   " "   " "    " "   " "    "*"   " "
## 2  ( 1 )  " "   " "  " "   " "  " " "*"   " "   " "    " "   " "    " "   "*"
## 3  ( 1 )  " "   " "  " "   " "  " " "*"   " "   "*"    "*"   " "    " "   " "
## 4  ( 1 )  " "   " "  " "   " "  " " "*"   " "   "*"    "*"   "*"    " "   " "
## 5  ( 1 )  " "   " "  " "   " "  " " "*"   " "   "*"    "*"   "*"    " "   " "
## 6  ( 1 )  " "   " "  " "   " "  " " "*"   " "   "*"    "*"   "*"    " "   " "
## 7  ( 1 )  " "   " "  " "   " "  " " "*"   " "   "*"    "*"   "*"    " "   " "
## 8  ( 1 )  " "   " "  " "   " "  " " "*"   " "   "*"    "*"   "*"    " "   " "
## 9  ( 1 )  " "   " "  " "   " "  " " "*"   " "   "*"    "*"   "*"    " "   " "
## 10  ( 1 ) "*"   "*"  " "   " "  " " "*"   " "   "*"    " "   " "    "*"   "*"
## 11  ( 1 ) "*"   "*"  " "   " "  " " "*"   " "   "*"    " "   " "    "*"   "*"
## 12  ( 1 ) "*"   "*"  " "   " "  " " "*"   " "   "*"    " "   " "    "*"   "*"
## 13  ( 1 ) "*"   "*"  " "   " "  " " "*"   "*"   "*"    " "   " "    "*"   "*"
## 14  ( 1 ) "*"   "*"  " "   "*"  " " "*"   "*"   "*"    " "   " "    "*"   "*"
## 15  ( 1 ) "*"   "*"  " "   " "  " " "*"   "*"   "*"    "*"   "*"    "*"   "*"
## 16  ( 1 ) "*"   "*"  " "   "*"  " " "*"   "*"   "*"    "*"   "*"    "*"   "*"
## 17  ( 1 ) "*"   "*"  "*"   " "  "*" "*"   "*"   "*"    "*"   "*"    "*"   "*"
## 18  ( 1 ) "*"   "*"  "*"   "*"  "*" "*"   "*"   "*"    "*"   "*"    "*"   "*"
## 19  ( 1 ) "*"   "*"  "*"   "*"  "*" "*"   "*"   "*"    "*"   "*"    "*"   "*"
##           CWalks LeagueN DivisionW PutOuts Assists Errors NewLeagueN
## 1  ( 1 )  " "    " "     " "       " "     " "     " "    " "
## 2  ( 1 )  " "    " "     " "       " "     " "     " "    " "
## 3  ( 1 )  " "    " "     " "       " "     " "     " "    " "
## 4  ( 1 )  " "    " "     " "       " "     " "     " "    " "
## 5  ( 1 )  " "    " "     "*"       " "     " "     " "    " "
## 6  ( 1 )  " "    " "     "*"       "*"     " "     " "    " "
## 7  ( 1 )  " "    " "     "*"       "*"     "*"     " "    " "
## 8  ( 1 )  " "    " "     "*"       "*"     "*"     "*"    " "
## 9  ( 1 )  " "    " "     "*"       "*"     "*"     "*"    "*"
## 10  ( 1 ) "*"    " "     "*"       "*"     "*"     " "    " "
## 11  ( 1 ) "*"    " "     "*"       "*"     "*"     " "    "*"
## 12  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 13  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 14  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 15  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 16  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 17  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 18  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 19  ( 1 ) "*"    "*"     "*"       "*"     "*"     "*"    "*"
```

```r
# Best model as per maximum Adjusted R^2:
coef(reg1,which.max(reg1_s$adjr2))
```

```
##  (Intercept)        AtBat         Hits        Walks       CAtBat        CRuns
##  154.1594365   -1.8722264    5.7547106    6.3852096   -0.2092154    1.7712871
##         CRBI       CWalks    DivisionW      PutOuts      Assists       Errors
```

```
##    1.1419024   -0.8989348 -121.8633686    0.2013298    0.5363804   -6.4726568
##   NewLeagueN
##   75.8378982
```

```r
# Best model as per minimum BIC (Schwartz's information criterion):
coef(reg1,which.min(reg1_s$bic))
```

```
##  (Intercept)        Walks       CAtBat        CHits       CHmRun    DivisionW
##  149.5853608    5.9582020   -0.5241611    2.0687418    1.7652811 -129.0165012
```

```r
# Best model as per minimum Cp (Mallows' Cp):
coef(reg1,which.min(reg1_s$cp))
```

```
##  (Intercept)        AtBat         Hits        Walks       CAtBat        CRuns
##  134.1593585   -1.9328699    5.8271814    6.5037245   -0.2063323    1.7820180
##         CRBI       CWalks    DivisionW      PutOuts      Assists   NewLeagueN
##    1.1120229   -0.8879140 -121.2135304    0.1925628    0.3316973   71.1433129
```

4. Find the training MSE and test MSE for each of the best models selected according to different criterion as mentioned in 3 and compare their values.

```r
## As per maximum Adjusted R^2:

#Best model:
model1=lm(Salary~(AtBat+Hits+Walks+CAtBat+CRuns+CRBI+CWalks+Division+PutOuts+Assists+Errors+NewLeague),

#Estimated response based on training data:
Y1_train=predict(model1,train_data)
#MSE based on Training Data
train_mse_1=mean((train_data$Salary-Y1_train)^2)

#Estimated response based on test data:
Y1_test=predict(model1,test_data)
#MSE based on Test Data
test_mse_1=mean((test_data$Salary-Y1_test)^2)

#-----------------------------------------------------------------------------------------------------------

## As per minimum BIC (Schwartz's information criterion):

#Best model:
model2=lm(Salary~(Walks+CAtBat+CHits+CHmRun+Division),train_data)

#Estimated response based on training data:
Y2_train=predict(model2,train_data)
#MSE based on Training Data
train_mse_2=mean((train_data$Salary-Y2_train)^2)

#Estimated response based on test data:
Y2_test=predict(model2,test_data)
#MSE based on Test Data
test_mse_2=mean((test_data$Salary-Y2_test)^2)
```

```r
#-----------------------------------------------------------------------------------------------------------

## As per minimum Cp (Mallows' Cp):

#Best model:
model3=lm(Salary~(AtBat+Hits+Walks+CAtBat+CRuns+CRBI+CWalks+Division),train_data)

#Estimated response based on training data:
Y3_train=predict(model3,train_data)
#MSE based on Training Data
train_mse_3=mean((train_data$Salary-Y3_train)^2)

#Estimated response based on test data:
Y3_test=predict(model3,test_data)
#MSE based on Test Data
test_mse_3=mean((test_data$Salary-Y3_test)^2)

#-----------------------------------------------------------------------------------------------------------

#Measures-
Measures_Best=c("Adjusted_R^2","BIC","Cp")

#Training MSEs-
Training_MSE_Best=c(train_mse_1,train_mse_2,train_mse_3)

#Test MSEs-
Test_MSE_Best=c(test_mse_1,test_mse_2,test_mse_3)

#Conclusion Table-
Conclusion_Best_Subset=data.frame(Measures_Best,Training_MSE_Best,Test_MSE_Best);Conclusion_Best_Subset
```

```
##   Measures_Best Training_MSE_Best Test_MSE_Best
## 1  Adjusted_R^2          88935.80      130066.9
## 2           BIC          97026.43      153090.5
## 3            Cp          95023.57      134013.3
```

Comment: The fitted model selected as per highest Adjusted R^2 value gives the least training & test MSEs given by-

Estimated_Salary=154.1594365-1.8722264*AtBat+5.7547106*Hits+6.3852096*Walks

-0.2092154*CAtBat+1.7712871*CRuns+1.1419024*CRBI-0.8989348*CWalks-121.8633686*Division

+0.2013298*PutOuts+0.5363804*Assists-6.4726568*Errors+75.8378982*NewLeague

5. Repeat 3-4 using forward stepwise selection method.

```r
## Forward Stepwise Subset Selection on the Training Data:

#leaps package required
library(leaps)

# regsubsets() with specified method "forward".
# nvmax fits the maximum no. of predictors considered in the full model
reg2=regsubsets(Salary~.,data=train_data,nvmax=19,method="forward");reg2
```

```
## Subset selection object
## Call: regsubsets.formula(Salary ~ ., data = train_data, nvmax = 19,
##     method = "forward")
## 19 Variables  (and intercept)
##            Forced in Forced out
## AtBat          FALSE      FALSE
## Hits           FALSE      FALSE
## HmRun          FALSE      FALSE
## Runs           FALSE      FALSE
## RBI            FALSE      FALSE
## Walks          FALSE      FALSE
## Years          FALSE      FALSE
## CAtBat         FALSE      FALSE
## CHits          FALSE      FALSE
## CHmRun         FALSE      FALSE
## CRuns          FALSE      FALSE
## CRBI           FALSE      FALSE
## CWalks         FALSE      FALSE
## LeagueN        FALSE      FALSE
## DivisionW      FALSE      FALSE
## PutOuts        FALSE      FALSE
## Assists        FALSE      FALSE
## Errors         FALSE      FALSE
## NewLeagueN     FALSE      FALSE
## 1 subsets of each size up to 19
## Selection Algorithm: forward
```

```r
#(*) in summary indicate the predictors in the best fitted subset model.
reg2_s=summary(reg2);reg1_s
```

```
## Subset selection object
## Call: regsubsets.formula(Salary ~ ., data = train_data, nvmax = 19)
## 19 Variables  (and intercept)
##            Forced in Forced out
## AtBat          FALSE      FALSE
## Hits           FALSE      FALSE
## HmRun          FALSE      FALSE
## Runs           FALSE      FALSE
## RBI            FALSE      FALSE
## Walks          FALSE      FALSE
## Years          FALSE      FALSE
## CAtBat         FALSE      FALSE
## CHits          FALSE      FALSE
## CHmRun         FALSE      FALSE
## CRuns          FALSE      FALSE
## CRBI           FALSE      FALSE
## CWalks         FALSE      FALSE
## LeagueN        FALSE      FALSE
## DivisionW      FALSE      FALSE
## PutOuts        FALSE      FALSE
## Assists        FALSE      FALSE
## Errors         FALSE      FALSE
## NewLeagueN     FALSE      FALSE
## 1 subsets of each size up to 19
```

```
## Selection Algorithm: exhaustive
##           AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits CHmRun CRuns CRBI
## 1  ( 1 )  " "   " "  " "   " "  " " " "   " "   " "    " "   " "    "*"   " "
## 2  ( 1 )  " "   " "  " "   " "  " " "*"   " "   " "    " "   " "    " "   "*"
## 3  ( 1 )  " "   " "  " "   " "  " " "*"   " "   "*"    "*"   " "    " "   " "
## 4  ( 1 )  " "   " "  " "   " "  " " "*"   " "   "*"    "*"   "*"    " "   " "
## 5  ( 1 )  " "   " "  " "   " "  " " "*"   " "   "*"    "*"   "*"    " "   " "
## 6  ( 1 )  " "   " "  " "   " "  " " "*"   " "   "*"    "*"   "*"    " "   " "
## 7  ( 1 )  " "   " "  " "   " "  " " "*"   " "   "*"    "*"   "*"    " "   " "
## 8  ( 1 )  " "   " "  " "   " "  " " "*"   " "   "*"    "*"   "*"    " "   " "
## 9  ( 1 )  " "   " "  " "   " "  " " "*"   " "   "*"    "*"   "*"    " "   " "
## 10  ( 1 ) "*"   "*"  " "   " "  " " "*"   " "   "*"    " "   " "    "*"   "*"
## 11  ( 1 ) "*"   "*"  " "   " "  " " "*"   " "   "*"    " "   " "    "*"   "*"
## 12  ( 1 ) "*"   "*"  " "   " "  " " "*"   " "   "*"    " "   " "    "*"   "*"
## 13  ( 1 ) "*"   "*"  " "   " "  " " "*"   "*"   "*"    " "   " "    "*"   "*"
## 14  ( 1 ) "*"   "*"  " "   "*"  " " "*"   "*"   "*"    " "   " "    "*"   "*"
## 15  ( 1 ) "*"   "*"  " "   " "  " " "*"   "*"   "*"    "*"   "*"    "*"   "*"
## 16  ( 1 ) "*"   "*"  " "   "*"  " " "*"   "*"   "*"    "*"   "*"    "*"   "*"
## 17  ( 1 ) "*"   "*"  "*"   " "  "*" "*"   "*"   "*"    "*"   "*"    "*"   "*"
## 18  ( 1 ) "*"   "*"  "*"   "*"  "*" "*"   "*"   "*"    "*"   "*"    "*"   "*"
## 19  ( 1 ) "*"   "*"  "*"   "*"  "*" "*"   "*"   "*"    "*"   "*"    "*"   "*"
##           CWalks LeagueN DivisionW PutOuts Assists Errors NewLeagueN
## 1  ( 1 )  " "    " "     " "       " "     " "     " "    " "
## 2  ( 1 )  " "    " "     " "       " "     " "     " "    " "
## 3  ( 1 )  " "    " "     " "       " "     " "     " "    " "
## 4  ( 1 )  " "    " "     " "       " "     " "     " "    " "
## 5  ( 1 )  " "    " "     "*"       " "     " "     " "    " "
## 6  ( 1 )  " "    " "     "*"       "*"     " "     " "    " "
## 7  ( 1 )  " "    " "     "*"       "*"     "*"     " "    " "
## 8  ( 1 )  " "    " "     "*"       "*"     "*"     "*"    " "
## 9  ( 1 )  " "    " "     "*"       "*"     "*"     "*"    "*"
## 10  ( 1 ) "*"    " "     "*"       "*"     "*"     " "    " "
## 11  ( 1 ) "*"    " "     "*"       "*"     "*"     " "    "*"
## 12  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 13  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 14  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 15  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 16  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 17  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 18  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 19  ( 1 ) "*"    "*"     "*"       "*"     "*"     "*"    "*"
```

```r
# Best model as per maximum Adjusted R^2:
coef(reg2,which.max(reg2_s$adjr2))
```

```
##  (Intercept)        AtBat         Hits        Walks        Years       CAtBat
##  207.8865937   -1.6603271    4.9552022    6.0893338  -12.4996617   -0.2504738
##        CHits        CRuns         CRBI       CWalks    DivisionW      PutOuts
##    0.3954684    1.4147636    1.0607231   -0.7225505 -123.4055931    0.1923965
##      Assists       Errors   NewLeagueN
##    0.5131110   -7.0115407   70.8774554
```

```r
# Best model as per minimum BIC (Schwartz's information criterion):
coef(reg2,which.min(reg2_s$bic))
```

```
##  (Intercept)        Walks       CAtBat        CHits        CRuns         CRBI
##  127.8159745    6.4746441   -0.4127407    1.1133040    1.0341530    0.8520825
##       CWalks    DivisionW
##   -0.5432536 -127.3877839
```

```r
# Best model as per minimum Cp (Mallows' Cp):
coef(reg2,which.min(reg2_s$cp))
```

```
##  (Intercept)        Walks       CAtBat        CHits        CRuns         CRBI
##  110.1015291    5.2753942   -0.4498891    1.0510348    1.3425171    0.9381944
##       CWalks    DivisionW      PutOuts      Assists       Errors   NewLeagueN
##   -0.5426157 -125.5159872    0.1609710    0.4662319   -7.6994411   74.6774479
```

```r
## As per maximum Adjusted R^2:

#Best model:
model21=lm(Salary~(AtBat+Hits+Walks+Years+CAtBat+CHits+CRuns+CRBI+CWalks+Division+PutOuts+Assists+Error

#Estimated response based on training data:
Y21_train=predict(model21,train_data)
#MSE based on Training Data
train_mse_21=mean((train_data$Salary-Y21_train)^2)

#Estimated response based on test data:
Y21_test=predict(model21,test_data)
#MSE based on Test Data
test_mse_21=mean((test_data$Salary-Y21_test)^2)

#-----------------------------------------------------------------------------------------------------------

## As per minimum BIC (Schwartz's information criterion):

#Best model:
model22=lm(Salary~(Walks+CAtBat+CHits+CRuns+CRBI+CWalks+Division),train_data)

#Estimated response based on training data:
Y22_train=predict(model22,train_data)
#MSE based on Training Data
train_mse_22=mean((train_data$Salary-Y22_train)^2)

#Estimated response based on test data:
Y22_test=predict(model22,test_data)
#MSE based on Test Data
test_mse_22=mean((test_data$Salary-Y22_test)^2)

#-----------------------------------------------------------------------------------------------------------

## As per minimum Cp (Mallows' Cp):
```

```r
#Best model:
model23=lm(Salary~(Walks+CAtBat+CHits+CRuns+CRBI+CWalks+Division+PutOuts+Assists+Errors+NewLeague),trai

#Estimated response based on training data:
Y23_train=predict(model23,train_data)
#MSE based on Training Data
train_mse_23=mean((train_data$Salary-Y23_train)^2)

#Estimated response based on test data:
Y23_test=predict(model23,test_data)
#MSE based on Test Data
test_mse_23=mean((test_data$Salary-Y23_test)^2)

#-----------------------------------------------------------------------------------------------------------

#Measures-
Measures_Forward=c("Adjusted_R^2","BIC","Cp")

#Training MSEs-
Training_MSE_Forward=c(train_mse_21,train_mse_22,train_mse_23)

#Test MSEs-
Test_MSE_Forward=c(test_mse_21,test_mse_22,test_mse_23)

#Conclusion Table-
Conclusion_Forward=data.frame(Measures_Forward,Training_MSE_Forward,Test_MSE_Forward);Conclusion_Forwar
```

```
##   Measures_Forward Training_MSE_Forward Test_MSE_Forward
## 1     Adjusted_R^2             88287.08         134001.7
## 2              BIC             95658.81         146540.9
## 3               Cp             90888.44         145923.9
```

Comment: The fitted model selected as per highest Adjusted R^2 value gives the least training & test MSEs given by-

Estimated_Salary=207.8865937-1.6603271*AtBat+4.9552022*Hits+6.0893338*Walks

-12.4996617*Years-0.2504738*CAtBat+0.3954684*CHits+1.4147636*CRuns

+1.0607231*CRBI-0.7225505*CWalks-123.4055931*Division+0.1923965*PutOuts

+0.5131110*Assists-7.0115407*Errors+70.8774554*NewLeague

6. Repeat 3-4 using backward stepwise selection method.

```r
## Backward Stepwise Subset Selection on the Training Data:

#leaps package required
library(leaps)

# regsubsets() with with specified method "backward".
# nvmax fits the maximum no. of predictors considered in the full model
reg3=regsubsets(Salary~.,data=train_data,nvmax=19,method="backward");reg3
```

```
## Subset selection object
```

```
## Call: regsubsets.formula(Salary ~ ., data = train_data, nvmax = 19,
##     method = "backward")
## 19 Variables  (and intercept)
##            Forced in Forced out
## AtBat          FALSE      FALSE
## Hits           FALSE      FALSE
## HmRun          FALSE      FALSE
## Runs           FALSE      FALSE
## RBI            FALSE      FALSE
## Walks          FALSE      FALSE
## Years          FALSE      FALSE
## CAtBat         FALSE      FALSE
## CHits          FALSE      FALSE
## CHmRun         FALSE      FALSE
## CRuns          FALSE      FALSE
## CRBI           FALSE      FALSE
## CWalks         FALSE      FALSE
## LeagueN        FALSE      FALSE
## DivisionW      FALSE      FALSE
## PutOuts        FALSE      FALSE
## Assists        FALSE      FALSE
## Errors         FALSE      FALSE
## NewLeagueN     FALSE      FALSE
## 1 subsets of each size up to 19
## Selection Algorithm: backward
```

```r
#(*) in summary indicate the predictors in the best fitted subset model.
reg3_s=summary(reg3);reg3_s
```

```
## Subset selection object
## Call: regsubsets.formula(Salary ~ ., data = train_data, nvmax = 19,
##     method = "backward")
## 19 Variables  (and intercept)
##            Forced in Forced out
## AtBat          FALSE      FALSE
## Hits           FALSE      FALSE
## HmRun          FALSE      FALSE
## Runs           FALSE      FALSE
## RBI            FALSE      FALSE
## Walks          FALSE      FALSE
## Years          FALSE      FALSE
## CAtBat         FALSE      FALSE
## CHits          FALSE      FALSE
## CHmRun         FALSE      FALSE
## CRuns          FALSE      FALSE
## CRBI           FALSE      FALSE
## CWalks         FALSE      FALSE
## LeagueN        FALSE      FALSE
## DivisionW      FALSE      FALSE
## PutOuts        FALSE      FALSE
## Assists        FALSE      FALSE
## Errors         FALSE      FALSE
## NewLeagueN     FALSE      FALSE
## 1 subsets of each size up to 19
```

```
## Selection Algorithm: backward
##           AtBat Hits HmRun Runs RBI Walks Years CAtBat CHits CHmRun CRuns CRBI
## 1  ( 1 )  " "   " "  " "   " "  " " " "   " "   " "    "*"   " "    " "   " "
## 2  ( 1 )  " "   " "  " "   " "  " " "*"   " "   " "    "*"   " "    " "   " "
## 3  ( 1 )  " "   " "  " "   " "  " " "*"   " "   "*"    "*"   " "    " "   " "
## 4  ( 1 )  " "   " "  " "   " "  " " "*"   " "   "*"    "*"   "*"    " "   " "
## 5  ( 1 )  " "   " "  " "   " "  " " "*"   " "   "*"    "*"   "*"    " "   " "
## 6  ( 1 )  " "   " "  " "   " "  " " "*"   " "   "*"    "*"   "*"    " "   " "
## 7  ( 1 )  " "   "*"  " "   " "  " " "*"   " "   "*"    "*"   "*"    " "   " "
## 8  ( 1 )  "*"   "*"  " "   " "  " " "*"   " "   "*"    "*"   "*"    " "   " "
## 9  ( 1 )  "*"   "*"  " "   " "  " " "*"   " "   "*"    "*"   "*"    " "   " "
## 10  ( 1 ) "*"   "*"  " "   " "  " " "*"   " "   "*"    "*"   "*"    " "   " "
## 11  ( 1 ) "*"   "*"  " "   " "  " " "*"   " "   "*"    "*"   "*"    "*"   " "
## 12  ( 1 ) "*"   "*"  " "   " "  " " "*"   " "   "*"    "*"   "*"    "*"   " "
## 13  ( 1 ) "*"   "*"  " "   " "  " " "*"   " "   "*"    "*"   "*"    "*"   " "
## 14  ( 1 ) "*"   "*"  " "   " "  " " "*"   "*"   "*"    "*"   "*"    "*"   " "
## 15  ( 1 ) "*"   "*"  " "   " "  "*" "*"   "*"   "*"    "*"   "*"    "*"   " "
## 16  ( 1 ) "*"   "*"  "*"   " "  "*" "*"   "*"   "*"    "*"   "*"    "*"   " "
## 17  ( 1 ) "*"   "*"  "*"   " "  "*" "*"   "*"   "*"    "*"   "*"    "*"   "*"
## 18  ( 1 ) "*"   "*"  "*"   "*"  "*" "*"   "*"   "*"    "*"   "*"    "*"   "*"
## 19  ( 1 ) "*"   "*"  "*"   "*"  "*" "*"   "*"   "*"    "*"   "*"    "*"   "*"
##           CWalks LeagueN DivisionW PutOuts Assists Errors NewLeagueN
## 1  ( 1 )  " "    " "     " "       " "     " "     " "    " "
## 2  ( 1 )  " "    " "     " "       " "     " "     " "    " "
## 3  ( 1 )  " "    " "     " "       " "     " "     " "    " "
## 4  ( 1 )  " "    " "     " "       " "     " "     " "    " "
## 5  ( 1 )  " "    " "     "*"       " "     " "     " "    " "
## 6  ( 1 )  " "    " "     "*"       "*"     " "     " "    " "
## 7  ( 1 )  " "    " "     "*"       "*"     " "     " "    " "
## 8  ( 1 )  " "    " "     "*"       "*"     " "     " "    " "
## 9  ( 1 )  " "    " "     "*"       "*"     "*"     " "    " "
## 10  ( 1 ) "*"    " "     "*"       "*"     "*"     " "    " "
## 11  ( 1 ) "*"    " "     "*"       "*"     "*"     " "    " "
## 12  ( 1 ) "*"    " "     "*"       "*"     "*"     " "    "*"
## 13  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 14  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 15  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 16  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 17  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 18  ( 1 ) "*"    " "     "*"       "*"     "*"     "*"    "*"
## 19  ( 1 ) "*"    "*"     "*"       "*"     "*"     "*"    "*"
```

```r
# Best model as per maximum Adjusted R^2:
coef(reg3,which.max(reg3_s$adjr2))
```

```
##  (Intercept)        AtBat         Hits        Walks       CAtBat        CHits
##  151.3709120   -1.5393497    4.6939494    6.1305798   -0.3519984    1.1424820
##       CHmRun        CRuns       CWalks    DivisionW      PutOuts      Assists
##    2.3082206    0.9269527   -0.5968973 -120.5643234    0.1975004    0.5618678
##       Errors   NewLeagueN
##   -7.1940771   71.8060203
```

```r
# Best model as per minimum BIC (Schwartz's information criterion):
coef(reg3,which.min(reg3_s$bic))
```

```
##  (Intercept)        Walks       CAtBat        CHits       CHmRun    DivisionW
##  149.5853608    5.9582020   -0.5241611    2.0687418    1.7652811 -129.0165012
```

```r
# Best model as per minimum Cp (Mallows' Cp):
coef(reg3,which.min(reg3_s$cp))
```

```
##  (Intercept)        AtBat         Hits        Walks       CAtBat        CHits
##  151.3709120   -1.5393497    4.6939494    6.1305798   -0.3519984    1.1424820
##       CHmRun        CRuns       CWalks    DivisionW      PutOuts      Assists
##    2.3082206    0.9269527   -0.5968973 -120.5643234    0.1975004    0.5618678
##       Errors   NewLeagueN
##   -7.1940771   71.8060203
```

```r
## As per maximum Adjusted R^2:

#Best model:
model31=lm(Salary~(AtBat+Hits+Walks+CAtBat+CHits+CHmRun+CRuns+CWalks+Division+PutOuts+Assists+Errors+Ne

#Estimated response based on training data:
Y31_train=predict(model31,train_data)
#MSE based on Training Data
train_mse_31=mean((train_data$Salary-Y31_train)^2)

#Estimated response based on test data:
Y31_test=predict(model31,test_data)
#MSE based on Test Data
test_mse_31=mean((test_data$Salary-Y31_test)^2)

#-----------------------------------------------------------------------------------------------------

## As per minimum BIC (Schwartz's information criterion):

#Best model:
model32=lm(Salary~(Walks+CAtBat+CHits+CHmRun+Division),train_data)

#Estimated response based on training data:
Y32_train=predict(model32,train_data)
#MSE based on Training Data
train_mse_32=mean((train_data$Salary-Y32_train)^2)

#Estimated response based on test data:
Y32_test=predict(model32,test_data)
#MSE based on Test Data
test_mse_32=mean((test_data$Salary-Y32_test)^2)

#-----------------------------------------------------------------------------------------------------

## As per minimum Cp (Mallows' Cp):
```

```r
#Best model:
model33=lm(Salary~(AtBat+Hits+Walks+CAtBat+CHits+CHmRun+CRuns+CWalks+Division+PutOuts+Assists+Errors+Ne

#Estimated response based on training data:
Y33_train=predict(model33,train_data)
#MSE based on Training Data
train_mse_33=mean((train_data$Salary-Y33_train)^2)

#Estimated response based on test data:
Y33_test=predict(model33,test_data)
#MSE based on Test Data
test_mse_33=mean((test_data$Salary-Y33_test)^2)

#-----------------------------------------------------------------------------------------------------------

#Measures-
Measures_Backward=c("Adjusted_R^2","BIC","Cp")

#Training MSEs-
Training_MSE_Backward=c(train_mse_31,train_mse_32,train_mse_33)

#Test MSEs-
Test_MSE_Backward=c(test_mse_31,test_mse_32,test_mse_33)

#Conclusion Table-
Conclusion_Backward=data.frame(Measures_Backward,Training_MSE_Backward,Test_MSE_Backward);Conclusion_Ba
```

```
##   Measures_Backward Training_MSE_Backward Test_MSE_Backward
## 1      Adjusted_R^2              88865.02          134736.7
## 2               BIC              97026.43          153090.5
## 3                Cp              88865.02          134736.7
```

Comment: The fitted model selected as per highest Adjusted R^2(& also lowest Cp) value gives the least training & test MSEs given by-

Estimated_Salary=151.3709120-1.5393497*AtBat+4.6939494*Hits+6.13057988*Walks-0.3519984*CAtBat

+1.1424820*CHits+2.3082206*CHumRun+0.9269527*CRuns-0.5968973*CWalks-120.5643234*Division

+0.1975004*PutOuts+0.5618678*Assists-7.1940771*Errors+71.8060203*NewLeague

7. Compare the results of the three methods in terms of choice of best model, train MSE amd test MSE.

```r
METHOD=c("Best_Subset","Forward_Stepwise","Backward_Stepwise")
TRAINING_MSE=c(Training_MSE_Best[1],Training_MSE_Forward[1],Training_MSE_Backward[1])
TEST_MSE=c(Test_MSE_Best[1],Test_MSE_Forward[1],Test_MSE_Backward[1])
Predictors_no=c(12,14,13)
Predictors=
CONCLUSION=data.frame(METHOD,TRAINING_MSE,TEST_MSE);CONCLUSION
```

```
##              METHOD TRAINING_MSE TEST_MSE
## 1       Best_Subset     88935.80 130066.9
## 2  Forward_Stepwise     88287.08 134001.7
## 3 Backward_Stepwise     88865.02 134736.7
```

Overall Comment: Here, for Best Subset selection training MSE & Test MSE combined are low.

Problem Set 7: Shrinkage Methods

Comparison of least squares, ridge regression, LASSO and Elastic Net

You would be working with Hitters data set available in the ISLR library. You have to predict the Salary based on all available variables. First make sure that you have deleted all rows with one or more missing values.

1. Split the data set into a training set and a test set randomly keeping 100 observations in test set and remaining in the training set.

```
# Ids of full data size
id=1:263

# 80% sample on ids of original data size for training sample
set.seed(123)
sample_id_new=sample(id,100)

# Training data
test_data_new=cleaned_data[sample_id_new,]

#Test data
train_data_new=cleaned_data[-sample_id_new,]
```

2. Train the model using least squares and report the train MSE and test MSE. ***Fitted Least Square Model:***

```
#Fitted Least Square Model
OLS_model=lm(Salary~.,data=train_data_new);OLS_model
```

```
##
## Call:
## lm(formula = Salary ~ ., data = train_data_new)
##
## Coefficients:
## (Intercept)        AtBat         Hits        HmRun         Runs          RBI
##   212.67449     -2.04895      5.85495      5.43622     -2.49915      2.32756
##       Walks        Years       CAtBat        CHits       CHmRun        CRuns
##     6.40986     -8.30051      0.01569     -0.54632     -2.38189      1.56320
##        CRBI       CWalks      LeagueN    DivisionW      PutOuts      Assists
##     1.26993     -0.95946     44.32184   -149.54532      0.41589      0.43589
##      Errors   NewLeagueN
##    -5.50162    -15.79562
```

Training MSE:

```
Salary_est_OLS_train=predict(OLS_model,data=train_data_new)
train_mse_OLS=mean((train_data_new$Salary-Salary_est_OLS_train)^2);train_mse_OLS
```

```
## [1] 105334.2
```

Test MSE:

```
Salary_est_OLS_test=predict(OLS_model,test_data_new)
test_mse_OLS=mean((test_data_new$Salary-Salary_est_OLS_test)^2);test_mse_OLS
```

```
## [1] 87803.91
```

3. Fit ridge regression models on the training set, with grid values of lambda. Report the following:

```
#Grid Values of lambda
lambda_grid=10^seq(5,-2,length=100)

#Fitting of Ridge Regression:
library("glmnet")
```

```
## Warning: package 'glmnet' was built under R version 4.5.3

## Loading required package: Matrix

## Warning: package 'Matrix' was built under R version 4.5.2

## Loaded glmnet 4.1-10
```

```
X_TRAIN=model.matrix(Salary~.,data=train_data_new)

Y_TRAIN=train_data_new$Salary

X_TEST=model.matrix(Salary~.,data=test_data_new)
Y_TEST=test_data_new$Salary

hitters_ridge=glmnet(X_TRAIN,Y_TRAIN,alpha=0,lambda=lambda_grid);hitters_ridge
```

```
##
## Call:  glmnet(x = X_TRAIN, y = Y_TRAIN, alpha = 0, lambda = lambda_grid)
##
##     Df  %Dev Lambda
## 1   19  2.03 100000
## 2   19  2.38  84980
## 3   19  2.77  72210
## 4   19  3.23  61360
## 5   19  3.75  52140
## 6   19  4.35  44310
## 7   19  5.04  37650
## 8   19  5.82  31990
## 9   19  6.70  27190
## 10  19  7.68  23100
## 11  19  8.78  19630
## 12  19  9.99  16680
## 13  19 11.32  14170
## 14  19 12.76  12050
```

```
## 15  19 14.29  10240
## 16  19 15.92   8697
## 17  19 17.61   7391
## 18  19 19.35   6280
## 19  19 21.12   5337
## 20  19 22.89   4535
## 21  19 24.62   3854
## 22  19 26.30   3275
## 23  19 27.91   2783
## 24  19 29.42   2364
## 25  19 30.83   2009
## 26  19 32.13   1707
## 27  19 33.32   1451
## 28  19 34.39   1233
## 29  19 35.37   1048
## 30  19 36.25    890
## 31  19 37.05    756
## 32  19 37.77    643
## 33  19 38.42    546
## 34  19 39.01    464
## 35  19 39.55    394
## 36  19 40.04    335
## 37  19 40.50    285
## 38  19 40.92    242
## 39  19 41.31    206
## 40  19 41.69    175
## 41  19 42.04    148
## 42  19 42.39    126
## 43  19 42.73    107
## 44  19 43.06     91
## 45  19 43.38     77
## 46  19 43.71     66
## 47  19 44.03     56
## 48  19 44.34     48
## 49  19 44.65     40
## 50  19 44.95     34
## 51  19 45.24     29
## 52  19 45.51     25
## 53  19 45.77     21
## 54  19 46.01     18
## 55  19 46.23     15
## 56  19 46.43     13
## 57  19 46.60     11
## 58  19 46.76      9
## 59  19 46.90      8
## 60  19 47.02      7
## 61  19 47.12      6
## 62  19 47.22      5
## 63  19 47.29      4
## 64  19 47.36      4
## 65  19 47.42      3
## 66  19 47.47      3
## 67  19 47.51      2
## 68  19 47.56      2
```

```
## 69  19 47.58      2
## 70  19 47.62      1
## 71  19 47.63      1
## 72  19 47.66      1
## 73  19 47.67      1
## 74  19 47.70      1
## 75  19 47.70      1
## 76  19 47.73      0
## 77  19 47.73      0
## 78  19 47.75      0
## 79  19 47.76      0
## 80  19 47.77      0
## 81  19 47.78      0
## 82  19 47.78      0
## 83  19 47.79      0
## 84  19 47.79      0
## 85  19 47.79      0
## 86  19 47.80      0
## 87  19 47.80      0
## 88  19 47.80      0
## 89  19 47.80      0
## 90  19 47.81      0
## 91  19 47.81      0
## 92  19 47.81      0
## 93  19 47.81      0
## 94  19 47.81      0
## 95  19 47.81      0
## 96  19 47.81      0
## 97  19 47.81      0
## 98  19 47.81      0
## 99  19 47.81      0
## 100 19 47.81      0
```

(i) the dimension of the coefficient matrix so obtained.

```r
dim(coef(hitters_ridge))
```

```
## [1]  21 100
```

(ii) The l2 norm of the regression coefficients corresponding to the 50th and 60th choice of lambda. Comment on the output.

```r
# Regression Coefficients w.r.t. 50th & 60th choice of lambda
coef_50=coef(hitters_ridge)[,50];coef_50
```

```
##   (Intercept)   (Intercept)         AtBat          Hits         HmRun
##  106.30930133    0.00000000   -0.56861594    1.57781724    1.45165595
##          Runs           RBI         Walks         Years        CAtBat
##    0.26148608    2.54343128    2.88812942   -4.15498762    0.01212119
##         CHits        CHmRun         CRuns          CRBI        CWalks
##    0.12212743    0.02430066    0.22110562    0.24061114   -0.27212940
##       LeagueN     DivisionW       PutOuts       Assists        Errors
```

```
##   54.66429035 -145.12121490    0.35444338    0.26953832   -5.05469505
##    NewLeagueN
##  -14.05949208
```

```
coef_60=coef(hitters_ridge)[,60];coef_60
```

```
##   (Intercept)   (Intercept)         AtBat          Hits         HmRun
##  1.755634e+02  0.000000e+00 -1.408336e+00  3.329840e+00  1.626042e+00
##          Runs           RBI         Walks         Years        CAtBat
## -1.917524e-01  3.107071e+00  4.844059e+00 -7.920993e+00 -5.315477e-03
##         CHits        CHmRun         CRuns          CRBI        CWalks
##  1.602586e-01 -3.013027e-01  4.667053e-01  3.938615e-01 -6.029094e-01
##       LeagueN     DivisionW       PutOuts       Assists        Errors
##  5.686263e+01 -1.526506e+02  3.918256e-01  3.813593e-01 -5.711395e+00
##    NewLeagueN
## -2.698770e+01
```

```
# Corresponding l2 norms
l2_norm=function(beta){sqrt(sum(beta^2))}
l2_norm_50=l2_norm(coef_50);l2_norm_50
```

```
## [1] 188.708
```

```
l2_norm_60=l2_norm(coef_60);l2_norm_60
```

```
## [1] 241.3126
```

(iv) Test MSE corresponding to these two choices of lambda.

```
predict_y_50=predict(hitters_ridge,s=lambda_grid[50],newx=X_TEST)
predict_y_60=predict(hitters_ridge,s=lambda_grid[60],newx=X_TEST)

test_mse_50=mean((Y_TEST-predict_y_50)^2);test_mse_50
```

```
## [1] 89317.28
```

```
test_mse_60=mean((Y_TEST-predict_y_60)^2);test_mse_60
```

```
## [1] 87580.18
```

Comment: As the tuning parameter lambda falls the l2 norm rises and the test mse falls.